

MANIKONDA THARUN

Full-Stack Engineer

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SUMMARY

Full-Stack Engineer with experience taking customer-facing software, internal operational tools, and applied AI systems from technical scope through deployment. Builds end-to-end applications with TypeScript, JavaScript, React, Node.js, Express.js, Java, PostgreSQL, and Redis, using REST APIs, testing, Docker, AWS, and CI/CD to support reliable releases. Recent work spans retrieval systems, real-time transaction validation, document processing, and reporting workflows. Translates requirements into technical scope, collaborates cross-functionally, and improves latency, consistency, and release predictability.

TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, SQL, Java.

Frontend Engineering: React, HTML5, CSS3, Responsive web development.

Backend Engineering: Node.js, Express.js, REST APIs, Spring Boot.

Data & Storage: PostgreSQL, Redis, Database indexing, SQL query tuning.

Cloud & Infrastructure: AWS, Docker, Amazon S3, Containerized deployments.

DevOps & CI/CD: Git, GitHub Actions, Docker Compose, CI workflows.

Testing & Quality: Jest, Cypress, Cross-browser testing, Automated test runs.

AI & LLM Systems: OpenAI API, Prompt engineering, AI ingestion pipelines, Retrieval systems.

System Design & Performance: Caching strategies, Horizontal scaling, Query optimization, Real-time processing.

PROFESSIONAL EXPERIENCE

McKinsey & Company | CA, USA

Full-Stack Engineer

May 2025 – Present

- Led enterprise delivery of applied AI workflows and customer-facing software, building ingestion and retrieval systems with Node.js, PostgreSQL, and OpenAI API under tight timelines to enable rapid client demos.
- Implemented ingestion pipelines using Node.js and AWS S3 with Postgres-backed metadata, applying prompt engineering to normalize inputs under data-quality constraints, resulting in consistently reliable retrieval performance for client pilots.
- Designed retrieval system integrating Redis caching and PostgreSQL indexes using TypeScript, optimizing query patterns under latency constraints and improving end-to-end response consistency for interactive customer demos.
- Built frontend React components and REST APIs with Express.js to surface LLM results, using Jest and Cypress for testing under usability requirements, improving client feedback cycles during workshops.
- Orchestrated containerized deployments with Docker and GitHub Actions, defining CI/CD pipelines and rollback strategies under enterprise compliance constraints to streamline releases for consulting engagements.
- Collaborated with cross-functional teams to translate client requirements into technical scope, using design reviews and backlog prioritization under delivery deadlines to maintain alignment and delivery predictability.

Uber | CA, USA

Full-Stack Engineer

February 2024 – May 2025

- Developed internal operational tooling and real-time workflows for transaction validation using Node.js, PostgreSQL, and Redis, coordinating with product teams under aggressive SLA expectations to improve operational efficiency.
- Implemented REST APIs in Express.js and TypeScript to validate transaction state, adding server-side checks and DB constraints under throughput constraints to reduce manual reconciliation across internal systems.
- Built React-based internal dashboards that surfaced validation results and alerts, using CSS3 and HTML5 under usability constraints to accelerate incident triage and stakeholder decision-making.
- Optimized caching strategies with Redis and query tuning in PostgreSQL, applying horizontal scaling considerations under load to stabilize real-time processing and reduce perceived latency for internal users.
- Maintained CI/CD workflows with GitHub Actions and Docker Compose, creating automated test runs with Jest under deployment constraints to ensure consistent releases to staging and production environments.

KPMG | India

Full-Stack Engineer

September 2021 – July 2022

- Led backend and reporting work on audit and compliance systems, developing Java backend services and document processing pipelines using PostgreSQL and Docker under strict data integrity constraints, improving report generation reliability.
- Implemented REST APIs in Java using Spring Boot and PostgreSQL under tight audit schema contracts, enabling automated document ingestion workflows and improving downstream reporting consistency.
- Optimized document processing pipeline using Redis caching, Docker containers, and multithreaded Java workers under throughput constraints across environments, reducing processing latency and stabilizing batch report exports.
- Built reporting workflows and SQL queries in PostgreSQL using Node.js scripts for data aggregation under compliance deadlines, producing audit-ready summaries consumed by internal audit teams.

- Collaborated with QA and DevOps using Git, Docker Compose, and CI pipelines to validate deployments under release controls, improving deployment repeatability and reducing manual rollback incidents.

Trigent Software | India

Frontend Developer

March 2020 – August 2021

- Led frontend engineering for UI migration projects, rebuilding responsive web delivery using React, HTML5, and CSS3 while coordinating QA-oriented implementation that improved cross-browser consistency and reduced visual regressions across devices.
- Implemented component library in React with TypeScript and CSS modules under tight release schedules, enabling reusable UI patterns and accelerating feature delivery while simplifying QA test coverage.
- Delivered responsive web pages using HTML5, CSS3, and JavaScript with cross-device testing under performance constraints, improving load behavior and meeting QA criteria for accessibility and responsiveness.

PROJECTS

Enterprise AI Document Assistant (RAG System)

- Architected a Retrieval-Augmented Generation (RAG) system using Python and LangChain to automate information extraction across 5,000+ enterprise documents, achieving 92% retrieval precision.
- Implemented a vector search pipeline using Pinecone with tuned chunking strategies (512-token segments with overlap), delivering semantic search responses in under 500ms.
- Leveraged GitHub Copilot and Cursor to develop FastAPI endpoints and React.js dashboards, reducing the end-to-end development cycle by 40%.
- Integrated AWS S3 for document storage, enabling semantic search results in under 500ms for complex queries.

NotchSafe — macOS Productivity Utility

- Engineered a native macOS utility using Swift and SwiftUI that transforms the MacBook notch into an interactive hub for encrypted storage and clipboard management.
- Integrated AES-256-GCM encryption and Touch ID biometric authentication, ensuring secure file handling with zero external dependencies.
- Designed a persistent clipboard monitoring system using NSEvent, resulting in a lightweight footprint (<5MB binary) and minimal system memory usage.

SmartHealth iOS: AI-Driven Wellness Tracker

- Developed a native iOS application using Swift and CoreML to provide real-time wellness insights based on user activity and behavioral data.
- Integrated a local NLP model for automated journaling and sentiment analysis of daily health logs.

EDUCATION

Master's in Computer Science

University of Alabama at Birmingham

August 2022 – December 2023

Bachelor of Engineering in Computer Science

K.S. Institute of Technology, Karnataka, India

June 2018 – May 2022

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate
- Certified MERN Developer – Namaste Dev
- Node.js Certified Developer – Coursera